

# Masked Wheel Generation Regression

Date: 2026-06-02

Test case: ivan-C1

## Branch Scope

This branch started as a Stage 2 mask-quality experiment:

```
detected wheel candidates -> VLM keep/reject filter -> cleaner wheel mask
```

After the mask filter looked useful, the scope expanded to a generation eval:

```
source car + target wheel reference + generated/filtered mask  
-> wheel replacement model
```

That means the branch now contains two related but separate questions:

1. Does the VLM filter improve the wheel mask?
2. If we pass that mask into generation/inpaint models, which model produces the best wheel replacement?

The pages below answer the second question for the current corrected rerun.

## Corrected Rerun

The previous frontier page included outputs generated with an old prompt that asked for a matte-black wheel while the actual reference image was silver. That made several rows visually misleading. The rerun below uses the corrected silver prompt:

```
Use the exact wheel design, color, finish, spoke pattern, center cap, and material from the reference image.  
Preserve original scene lighting.  
Do not alter car body, car color, background, road, tires, or unmasked pixels.
```

Cases:

- ivan-C1: daylight control case.

- `ivan-N1`: night case.
- `ivan-N2`: night wet-looking case. There is no explicit rain metadata in the local dataset, so this is not labeled as confirmed rain.

01. ivan-c1 (daylight control) / AITUNNEL gpt-image-1.5

AITUNNEL / OpenAI-compatible edits | corrected silver prompt | ref passed yes | mask passed yes

Input image: source car



Input image: wheel reference



Output / status



02. ivan-c1 (daylight control) / OpenAI gpt-image-2

Official OpenAI API (images/edits streamed) | corrected silver prompt | ref passed yes | mask passed yes

Input image: source car



Input image: wheel reference



Output / status



03. ivan-c1 (daylight control) / Gemini 3 Pro Image Edit corrected

fat.ai / gemini-3-pro-image-preview-edits | corrected silver prompt | ref passed yes | mask passed yes

Input image: source car



Input image: wheel reference



Output / status



04. ivan-c1 (daylight control) / Reve Direct masked corrected

Reve API / image remix | corrected silver prompt | ref passed yes | mask passed yes

Input image: source car



Input image: wheel reference



Output / status



05. ivan-n1 (night) / AITUNNEL gpt-image-1.5

AITUNNEL / OpenAI-compatible edits | corrected silver prompt | ref passed yes | mask passed yes

Input image: source car



Input image: wheel reference



Output / status



06. ivan-n1 (night) / OpenAI gpt-image-2

Official OpenAI API (images/edits streamed) | corrected silver prompt | ref passed yes | mask passed yes

Input image: source car



Input image: wheel reference



Output / status



07. ivan-n1 (night) / Gemini 3 Pro Image Edit corrected

fal.ai / gemini-3-pro-image-preview/edit | corrected silver prompt | ref passed yes | mask passed yes

Input image: source car



Input image: wheel reference



Output / status



08. ivan-n1 (night) / ReVe Direct masked corrected

ReVe API / image remix | corrected silver prompt | ref passed yes | mask passed yes

Input image: source car



Input image: wheel reference



Output / status



09. ivan-n2 (night wet-looking) / AITUNNEL gpt-image-1.5

AITUNNEL / OpenAI-compatible edits | corrected silver prompt | ref passed yes | mask passed yes

Input image: source car



Input image: wheel reference



Output / status



10. ivan-n2 (night wet-looking) / OpenAI gpt-image-2

Official OpenAI API (images/edits streamed) | corrected silver prompt | ref passed yes | mask passed yes

Input image: source car



Input image: wheel reference



Output / status



11. ivan-n2 (night wet-looking) / Gemini 3 Pro Image Edit corrected

fal.ai / gemini-3-pro-image-preview-edits | corrected silver prompt | ref passed yes | mask passed yes

Input image: source car



Input image: wheel reference



Output / status



12. ivan-n2 (night wet-looking) / ReVe Direct masked corrected

Reve API / image remix | corrected silver prompt | ref passed yes | mask passed yes

Input image: source car



Input image: wheel reference



Output / status



Corrected rerun status:

| Model                                   | Cases      | Status                               | Read                                                                                                 |
|-----------------------------------------|------------|--------------------------------------|------------------------------------------------------------------------------------------------------|
| gpt - image - 1 . 5 via AITUNNEL        | C1, N1, N2 | 3/3 completed after one N2 retry     | best current OpenAI-compatible fallback through AITUNNEL                                             |
| gpt - image - 2 via official OpenAI API | C1, N1, N2 | 3/3 completed with streaming default | official masked wheel-edit baseline; now added into the same visual comparison pages                 |
| Gemini 3 Pro Image Edit via fal.ai      | C1, N1, N2 | 3/3 completed                        | no longer black, but often fails the edit task by cropping or generating product-wheel style outputs |
| Reve Direct masked corrected            | C1, N1, N2 | 3/3 completed                        | stable masked baseline                                                                               |

gpt - image - 1 . 5 note: the first N2 request failed with Server disconnected without sending a response; a single retry completed. This means AITUNNEL edits are usable but still have transport-level instability.

## Direct OpenAI GPT Image 2 Baseline

After the AITUNNEL gpt - image - 2 failures, the same masked wheel-copy path was retested against the official OpenAI API using the same corrected silver reference prompt and the same image[] + image[] + mask request shape.

What was tested:

- non-stream gpt - image - 2 on ivan - C1, quality=medium: failed with Server disconnected without sending a response
- non-stream gpt - image - 2 on ivan - C1, quality=low: failed with Server disconnected without sending a response
- stream-enabled gpt - image - 2 on corrected local cases ivan - C1, ivan - N1, ivan - N2, quality=low: 3/3 completed

Direct OpenAI read:

| Model                                   | Cases                | Status                               | Read                                                                            |
|-----------------------------------------|----------------------|--------------------------------------|---------------------------------------------------------------------------------|
| gpt - image - 2 via official OpenAI API | C1, N1, N2           | 3/3 completed with streaming default | official baseline now confirmed for source image + wheel reference + alpha mask |
| gpt - image - 2 via official OpenAI API | C1 non-stream checks | 0/2 completed                        | plain non-stream response path still disconnected in this runner                |

Interpretation:

- gpt - image - 2 is API-accessible and does support the required wheel-edit workflow on the official OpenAI endpoint.
- The earlier blocker was not model availability.
- In this repo's current runner, streaming is the stable transport path for direct OpenAI image edits.
- Those successful direct OpenAI outputs are now inserted into the same corrected rerun comparison pages as the AITUNNEL, Gemini, and Reve examples.

## Rain Stress Cases

Additional user-provided images were tested with the same corrected silver prompt and manually prepared wheel masks.

Cases:

- rain-flood-bmw: flooded road / water splash.
- night-heavy-rain-sedan: night heavy rain.
- istock-rain-splash: rain splash with visible stock watermark; this is a stress-test only, not a presentation-quality source.



### 01. rain-flood-bmw (flooded road / water splash) / AITUNNEL gpt-image-1.5

AITUNNEL / OpenAI-compatible edits | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



### 02. rain-flood-bmw (flooded road / water splash) / Gemini 3 Pro Image Edit corrected

fal.ai / gemini-3-pro-image-preview/edit | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



### 03. rain-flood-bmw (flooded road / water splash) / Reve Direct masked corrected

Reve API / image remix | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



04. night-heavy-rain-sedan (night heavy rain) / AITUNNEL gpt-image-1.5

AITUNNEL / OpenAI-compatible edits [corrected silver prompt] ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status

FAILED

failed twice: Server disconnected without sending a response.

05. night-heavy-rain-sedan (night heavy rain) / Gemini 3 Pro Image Edit corrected

fal.ai / gemini-3-pro-image-preview/edit [corrected silver prompt] ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



06. night-heavy-rain-sedan (night heavy rain) / Reve Direct masked corrected

Reve API / image remix [corrected silver prompt] ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



#### 07. istock-rain-splash (rain splash / watermark stress-case) / AITUNNEL gpt-image-1.5

AITUNNEL / OpenAI-compatible edits | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



#### 08. istock-rain-splash (rain splash / watermark stress-case) / Gemini 3 Pro Image Edit corrected

fal.ai / gemini-3-pro-image-preview/edit | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



#### 09. istock-rain-splash (rain splash / watermark stress-case) / Reve Direct masked corrected

Reve API / image remix | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



Rain stress status:

| Model                              | Cases                                                                          | Status                                                      | Read                                                           |
|------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------|
| gpt - image - 1.5 via AITUNNEL     | rain - flood - bmw,<br>night - heavy - rain - sedan,<br>istock - rain - splash | 2/3 completed;<br>night - heavy - rain - sedan failed twice | useful when it completes, but transport<br>instability remains |
| Gemini 3 Pro Image Edit via fal.ai | all 3 rain cases                                                               | 3/3 completed                                               | still inconsistent as masked car editor                        |
| Reve Direct masked corrected       | all 3 rain cases                                                               | 3/3 completed                                               | most stable under these stress cases                           |

# Status Summary

Current corrected rerun:

| Model                                   | Cases             | Status                               | Result                                             |
|-----------------------------------------|-------------------|--------------------------------------|----------------------------------------------------|
| gpt - image - 1.5 via AITUNNEL          | C1, N1, N2        | completed after one N2 retry         | usable OpenAI-compatible fallback                  |
| gpt - image - 2 via official OpenAI API | C1, N1, N2        | completed with streaming default     | confirmed official baseline for masked wheel edits |
| Gemini 3 Pro Image Edit via fal.ai      | C1, N1, N2        | completed                            | still not reliable for masked car edits            |
| Reve Direct masked                      | C1, N1, N2        | completed                            | stable masked baseline                             |
| gpt - image - 1.5 via AITUNNEL          | rain stress cases | 2/3 completed; one case failed twice | transport instability remains                      |
| Gemini 3 Pro Image Edit via fal.ai      | rain stress cases | completed                            | visually inconsistent                              |
| Reve Direct masked                      | rain stress cases | completed                            | stable                                             |

AITUNNEL endpoint diagnostics:

| Model             | Endpoint                                       | Status                                                 |
|-------------------|------------------------------------------------|--------------------------------------------------------|
| gpt - image - 2   | /v1/images/generations                         | works                                                  |
| gpt - image - 2   | /v1/images/edits                               | fails: Server disconnected without sending a response. |
| gpt - image - 1.5 | /v1/images/edits with image[] + image[] + mask | works, but one transient disconnect on N2              |

|                  |                                                |                                                        |
|------------------|------------------------------------------------|--------------------------------------------------------|
| gpt-image-1-mini | /v1/images/edits with image[] + image[] + mask | works                                                  |
| gpt-image-1      | /v1/images/edits                               | fails: Server disconnected without sending a response. |

Official OpenAI endpoint diagnostics:

| Model       | Endpoint                               | Status                                                                   |
|-------------|----------------------------------------|--------------------------------------------------------------------------|
| gpt-image-2 | /v1/images/edits non-stream            | failed twice on ivan-C1: Server disconnected without sending a response. |
| gpt-image-2 | /v1/images/edits with streaming events | works for ivan-C1, ivan-C2, ivan-N2                                      |

Conclusion:

- AITUNNEL now supports the required image[] + mask request shape for gpt-image-1.5 and gpt-image-1-mini.
- gpt-image-2 still fails on /v1/images/edits, including simpler edit control requests.
- On the official OpenAI API, gpt-image-2 is a confirmed masked wheel-edit baseline when streaming is enabled.
- For the current wheel-copy eval, gpt-image-1.5 remains the usable AITUNNEL fallback while gpt-image-2 is tested directly on OpenAI.

## GPT Image 2 Decision

gpt-image-2 remains the main candidate to test as a potential Reve replacement because it should support the exact request shape required by this product:

```
source car + target wheel reference + alpha mask
```

Current AITUNNEL status:

- gpt-image-2 is visible in /v1/models.
- gpt-image-2 works for /v1/images/generations.
- gpt-image-2 still fails for /v1/images/edits.
- The failure is transport-level disconnect, not a JSON validation error.

Current official OpenAI status:

- `gpt-image-2` works for `/v1/images/edits` on the exact masked wheel-copy request shape used by this project.
- In this repo's runner, the stable direct path is streaming mode; two non-stream checks on `ivan-C1` disconnected before a response body arrived.

Operational decision:

- Do not block the model evaluation on `gpt-image-2` availability.
- Treat official OpenAI `gpt-image-2` as a confirmed baseline for masked wheel edits.
- Keep stream mode enabled by default for direct OpenAI image-edit runs in this repo.
- Use AITUNNEL `gpt-image-1.5` as the current OpenAI-compatible fallback for AITUNNEL demos and gateway regression checks.
- Do not call `gpt-image-1.5` a full Reve replacement yet; it needs more cases and stability checks.
- Keep asking AITUNNEL support to fix `/v1/images/edits` for `gpt-image-2`.

## Next Steps

- Run the fine-tuned real/fake wheel classification model on generated outputs to separate true wheel-copy results from visually plausible but incorrect wheel generations.
- Improve wheel masking quality before the next model comparison; current manual and filtered masks are good enough for stress testing, but mask edge quality still affects rain, splash, and night cases.

## Output Files

- Corrected rerun summary CSV: `docs/assets/aitunnel-gpt-image-regression/corrected-rerun-c1-n1-n2-summary.csv`
- Official OpenAI baseline summary CSV: `docs/assets/aitunnel-gpt-image-regression/official-openai-gpt-image-2-summary.csv`
- Rain stress summary CSV: `docs/assets/aitunnel-gpt-image-regression/rain-rerun-summary.csv`

## Raw Results

- `tmp/openai-image-edit-eval/results-aitunnel-gpt-image-15-corrected-c1-n1-n2/openai_image_edit_results.csv`
- `tmp/openai-image-edit-eval/results-aitunnel-gpt-image-15-corrected-n2-retry/openai_image_edit_results.csv`
- `tmp/openai-image-edit-eval/direct-openai-gpt-image-2-c1-medium-corrected/openai_image_edit_results.csv`
- `tmp/openai-image-edit-eval/direct-openai-gpt-image-2-c1-low-corrected/openai_image_edit_results.csv`
- `tmp/openai-image-edit-eval/direct-openai-gpt-image-2-corrected-local-3cases-stream-default/openai_image_edit_results.csv`
- `tmp/openai-image-edit-eval/direct-openai-gpt-image-2-n1-low-corrected-stream-default/openai_image_edit_results.csv`
- `tmp/fal-inpaint-eval/results-corrected-silver-gemini-c1-n1-n2/fal_inpaint_results.csv`
- `tmp/reve-image-edit-eval/results-corrected-silver-masked-c1-n1-n2/reve_image_edit_results.csv`
- `tmp/rain-wheel-eval/results-aitunnel-gpt-image-15/openai_image_edit_results.csv`

- tmp/rain-wheel-eval/results-aitunnel-gpt-image-15-night-heavy-rain-retry/openai\_image\_edit\_results.csv
- tmp/rain-wheel-eval/results-gemini/fal\_inpaint\_results.csv
- tmp/rain-wheel-eval/results-reve-masked/reve\_image\_edit\_results.csv